

Definition and first terms

$(4-1\,x)\,A(x)=3+A(x)^3$, with the origin-normalized solution.

Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+1*x*T+3*T^2+1*T^3$.

$a(0),\ldots,a(7)=1,1,4,29,263,29088,331749$.

Kernel used throughout: $\rho(u)=u\,\frac{1-3\,u^1-1\,u^2}{1+1\,u}$, so $x=\rho(T(x))$.

Typogeometry

1x1

{1,1}x4

{1,{1,1}}x16

{1,1,1}x1

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 3, Δ_2^{marked} : 1, Δ_3 : 1.

Coefficient and contour extraction

$$\mathcal{K}_{n,h}=\{(k_1,\ldots,k_2)\in\mathbb{Z}_{\geq 0}^2:h+\sum_{j=1}^2jk_j=n-1\},\quad K=\sum_{j=1}^2k_j,$$

$$a(n)=\frac{1}{n}\sum_{h=0}^{n-1}\binom{n}{h}\sum_{\boldsymbol{k}\in\mathcal{K}_{n,h}}\binom{n+K-1}{K}\binom{K}{k_1,\ldots,k_2}3^{k_1}1^{k_2}.$$

$$a(n)=\frac{1}{2\pi i n}\oint_\gamma\frac{du}{\rho(u)^n},\quad n\geq 1.$$

Recurrence

$$\sum_{r=0}^4P_r(n)a(n+r)=0,\,n\geq 1,\,\text{with}$$

$$\begin{aligned}P_0(n)&=4\,n^3+2\,n^2-2\,n\\P_1(n)&=-64\,n^3-168\,n^2-136\,n-32\\P_2(n)&=384\,n^3+1824\,n^2+2848\,n+1472\\P_3(n)&=-781\,n^3-5825\,n^2-14286\,n-11520\\P_4(n)&=52\,n^3+468\,n^2+1352\,n+1248\end{aligned}$$

Differential equation

$$\begin{aligned}(0)\,A(x)&+(4\,x^5-32\,x^4+64\,x^3)\,A'(x)\\&+(14\,x^6-168\,x^5+672\,x^4-1139\,x^3)\,A^{(2)}(x)+(4\,x^7-64\,x^6+384\,x^5-781\,x^4+52\,x^3)\,A^{(3)}(x)=0.\end{aligned}$$

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n,u,x)=N/D\text{ (exact 246-character numerator in embedded payload).}$$

$$\text{Telescoping identity: }\sum_rP_r(n)H_{n+r}(u)=\partial_u(R(n,u)H_n(u)).$$

Matrix dimensions: 6×6; remainder 2×3, rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.